

The Uniform Distribution

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Introduction

The uniform distribution spreads probability equally across its support. The continuous uniform on $[a, b]$ is the building block of random-number generation; the discrete uniform on $\{1, \dots, k\}$ describes fair dice and random selection. It is the simplest non-trivial distribution.

Prerequisites

Continuous and discrete random variables.

Theory

Continuous uniform $\text{Uniform}(a, b)$ has PDF

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b,$$

with moments $E[X] = (a+b)/2$ and $\text{Var}(X) = (b-a)^2/12$.

Discrete uniform on $\{x_1, \dots, x_k\}$: $P(X = x_i) = 1/k$ for each i . For $\{1, \dots, k\}$: $E[X] = (k+1)/2$, $\text{Var}(X) = (k^2 - 1)/12$.

Probability integral transform: if X has a continuous CDF F_X , then $F_X(X) \sim \text{Uniform}(0, 1)$. The inverse: $X = F_X^{-1}(U)$ where $U \sim \text{Uniform}(0, 1)$ generates X from uniform draws. This is the inverse-CDF method of simulation.

Assumptions

Continuous uniform assumes the variable genuinely has constant density; discrete uniform assumes genuinely equal probabilities across a specified set.

R Implementation

```
set.seed(2026)

# Continuous uniform
U <- runif(1e5, min = 0, max = 1)
c(mean = mean(U), theoretical = 0.5,
  var = var(U), theoretical_var = 1/12)

# Inverse-CDF method: generate exponential from uniform
lambda <- 2
E <- -log(1 - U) / lambda # F^{-1}(u) = -log(1 - u) / lambda
c(empirical_mean = mean(E), exponential_mean = 1 / lambda)

# Discrete uniform via sample()
```

```
die <- sample(1:6, 1e5, replace = TRUE)
c(mean_die = mean(die), theoretical = 3.5,
  var_die = var(die), theoretical_var = 35/12)
```

Output & Results

mean	theoretical	var	theoretical_var
0.5003	0.5000	0.0835	0.0833

empirical_mean	exponential_mean
0.4983	0.500

mean_die	theoretical	var_die	theoretical_var
3.504	3.500	2.924	2.917

Empirical moments match theoretical exactly up to Monte Carlo error, and the inverse-CDF method recovers the exponential mean.

Interpretation

Uniforms rarely appear as substantive models of real data but pervade simulation, randomisation, and Monte Carlo methods. Reporting “trial assignment was simulated via `sample()`” is implicitly invoking a discrete uniform.

Practical Tips

- `runif(n)` defaults to $[0, 1]$; specify min/max explicitly to avoid surprises.
- The uniform distribution has zero skewness and excess kurtosis -1.2 (light tails).
- For true randomness from hardware sources, avoid pseudo-random `runif`; use `/dev/urandom` or the `random` R package with online services.
- Uniform priors are often overly informative on transformed scales (a uniform on σ is not uniform on σ^2).
- The Jacobian of the inverse-CDF transform is $1/f_X(x)$; this is why the probability integral transform works.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes’ theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots